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FINAL REPORT

On the START Program

Automated analysis of nuclear emulsion exposed to ions $^{124}\text{Xe}^{+54}$ at about 300 MeV per nucleon at the JINR Nuclotron

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Abstract

The article presents results of an analysis of a nuclear emulsion exposed in March 2026 with an extracted beam of $^{124}\text{Xe}^{+54}$ ions at around 300 MeV per nucleon at the JINR Nuclotron. The primary objective was to determine the beam profile and its uniformity, and to locate the Bragg peak position for applied research in electronic components and radiation biology. Three stacks of 8–10 emulsion layers of 60 μm thickness each poured onto a 2 mm thick glass base measuring $9 \times 12 \text{ cm}^2$, were exposed transversely to the beam, allowing capture of the bulk of the beam and tracking of ion deceleration until complete stop or the formation of long-range fragments. Digital photography of approximately 80% of each layer area was performed using an intelligent microscope Olympus BX63, and the frames were combined into a complete layer image using the microscope's software. The high ionization of the ions made it sufficient to use a $4\times$ objective lens to detect cross-sectional images of their tracks. Counting and determination of the track parameters were performed using ImageJ. A Gaussian Mixture Model on grain areas resolves three particle classes corresponding to light fragments, mid-mass fragments, and heavy residues including the primary beam. The Bragg peak is observed at 8–10 mm (Stack01) and 10–12 mm (Stacks 02/03) depth.

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1 Accelerator, research goals and objectives

A long-term perspective at VBLHEP has been opened up with the NICA (Nuclotron-based Ion Collider fAcility) megaproject, whose primary scientific goal is the study of quark–gluon degrees of freedom in nuclear matter at maximum baryonic density. NICA provides extracted and colliding beams of heavy relativistic nuclei and polarized protons and deuterons at energies up to 4.5 GeV/nucleon for extracted beams and up to 11 GeV/nucleon in the collider mode.

1.1 Accelerator cascade

The NICA accelerator complex consists of several stages:

1. **Electron String Ion Source (ESIS)** and/or **KRION-6T**: Produce highly charged ions of the desired species (e.g. Xe^{+54} for this experiment).
2. **HILAC** (Heavy Ion Linear Accelerator): A new linear accelerator that accelerates ions to 3.2 MeV/nucleon for injection into the booster.
3. **Superconducting Booster Synchrotron**: Accumulates and accelerates ions to 600 MeV/nucleon before transfer to the Nuclotron.
4. **Nuclotron**: The main superconducting synchrotron, capable of accelerating heavy ions up to 4.5 GeV/nucleon. Beams can be slowly extracted over several seconds for fixed-target experiments or injected into the collider rings.
5. **Collider**: Two superconducting storage rings (each 503 m circumference) that bring counter-rotating beams into collision at interaction points for experiments such as MPD and SPD.

1.2 Beam extraction and the present irradiation

For the present experiment, $^{124}\text{Xe}^{+54}$ ions were accelerated by the Nuclotron to approximately 300 MeV/nucleon and slowly extracted via the beam extraction system to the VBLHEP experimental hall. The beam energy at the irradiation point was approximately 280 MeV/nucleon, as determined from the Nuclotron magnetic field settings. At this energy the ions have a range of approximately 10–15 mm in the emulsion–glass stack, allowing the majority of the beam to stop within the 8–10 plate assembly while still producing measurable signals in the downstream plates from long-range nuclear fragments.

The extracted beam was delivered in a series of macro-pulses (spills), each of several hundred milliseconds duration, with typical intensities of 10^4 – 10^5 ions per spill at the irradiation point. The three emulsion stacks were exposed with different numbers of cycles (1, 5, and 10 spills for Stacks 01, 02, and 03 respectively) to provide a range of fluences for statistical analysis.

1.3 Applied physics programme at NICA

Beyond its fundamental physics goals, the NICA accelerator complex supports an extensive applied research programme:

- **Radiation hardness of electronics:** Heavy-ion beams at energies of hundreds of MeV/nucleon are ideal for testing single-event effects (SEE) in microelectronics, simulating the space radiation environment. The xenon beam used here provides a well-characterised source of high-LET radiation.
- **Radiation biology:** The Bragg peak of heavy ions in tissue-like media is exploited for hadron therapy. Measurements of the range and fragmentation of ^{124}Xe in emulsion provide benchmark data for treatment planning codes.
- **Nuclear fragmentation:** The breakup of heavy projectiles in the emulsion produces a spectrum of secondary fragments whose yields and angular distributions are relevant for both fundamental nuclear physics and space radiation shielding.
- **Detector development:** Nuclear emulsion is itself a candidate detector for the NICA SPD (Spin Physics Detector) and for future fixed-target experiments, motivating quantitative studies of its response to heavy ions.

The present work directly addresses the first point (beam characterisation for electronics testing) while contributing to the second and third by measuring the Bragg curve and identifying fragment populations.

The specific goals of this work are:

1. To determine the beam profile and uniformity of the extracted ^{124}Xe beam at the irradiation point;
2. To measure the range and locate the Bragg peak of 280 MeV/nucleon xenon ions in the emulsion-glass stack;
3. To classify the nuclear fragmentation products by their ionization density using unsupervised machine learning;
4. To compare the experimental range with SRIM simulations.

These results are essential for the calibration of nuclear emulsion as a detector for heavy-ion fragmentation studies, and for applications in radiation hardness testing of electronic components and in radiobiology.

2 Nuclear emulsion in brief

Nuclear emulsion is a particle detector consisting of a photographic gelatine layer embedded with silver bromide (AgBr) crystals. When a charged particle passes through the emulsion, it ionises the medium along its trajectory. During chemical development, the latent image centres in the AgBr crystals along the particle's path are reduced to metallic silver grains, forming a visible track that can be observed under an optical microscope. The analysis presented here is part of the BECQUEREL project at LHEP JINR [4, 5].

The principal advantage of nuclear emulsion is its sub-micrometre spatial resolution, which makes it uniquely suited for measuring the interaction vertices and ranges of heavy ions in the few-hundred MeV/nucleon energy regime. Unlike electronic detectors, emulsion integrates the exposure passively, requires no power or readout during irradiation, and provides a permanent record of every particle trajectory within its volume.

2.1 Energy loss in emulsion

The energy loss of a heavy ion in emulsion follows the Bethe–Bloch formula [3]:

$$-\frac{dE}{dx} = \frac{4\pi}{m_e c^2} \cdot \frac{n_e z^2}{\beta^2} \cdot \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left[\ln \frac{2m_e c^2 \beta^2}{I(1 - \beta^2)} - \beta^2 \right],$$

where z is the projectile charge, $\beta = v/c$ its velocity, n_e the electron density of the medium, and I the mean excitation potential. The grain size in the developed emulsion is proportional to the local energy deposition, making it possible to distinguish particle types by their ionization density. For a ^{124}Xe ion at 280 MeV/nucleon ($\beta \approx 0.63$), the ionization is far above the minimum, producing large, easily detectable grain clusters.

2.2 Emulsion composition

The emulsion layer consists of:

- AgBr microcrystals ($\sim 40\%$ by volume, density 6.47 g cm^{-3});
- Gelatine binder ($\sim 60\%$ by volume, density 1.16 g cm^{-3});
- Bulk density approximately 3.3 g cm^{-3} .

After exposure, the plates are chemically developed, fixed, and washed following standard nuclear emulsion protocols.

3 Stack description

Three stacks of nuclear emulsion plates were exposed to an extracted beam of $^{124}\text{Xe}^{+54}$ ions in March 2026 at the JINR Nuclotron.

3.1 Plate structure

Each plate consists of a $60\text{ }\mu\text{m}$ layer of nuclear emulsion poured onto a 2 mm thick glass substrate. The plates measure approximately $9 \times 12\text{ cm}^2$. The emulsion layer is single-sided (one sensitive layer per plate, no backing emulsion).

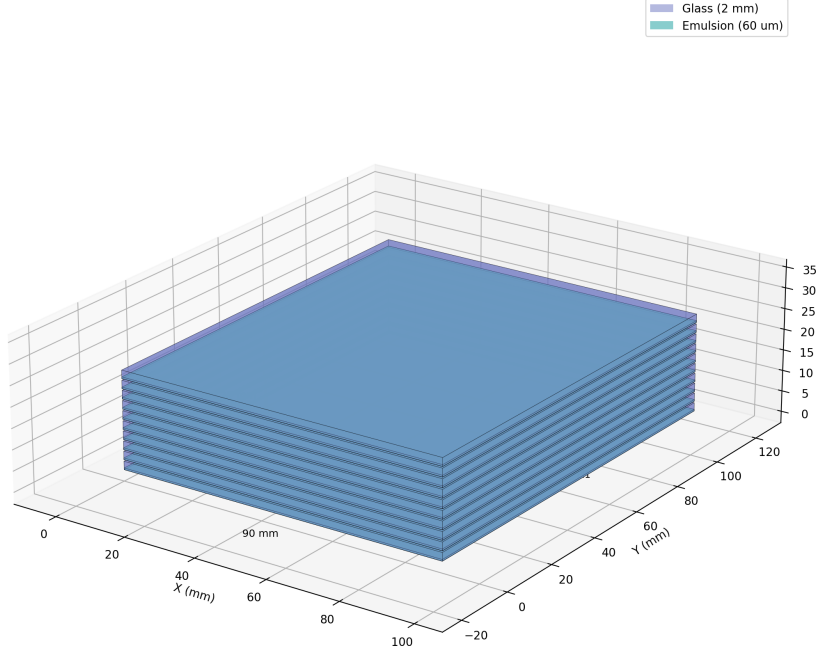


Figure 1: 3D model of the emulsion stack showing 10 plates stacked perpendicular to the beam direction (red arrow). Each plate consists of a $60\text{ }\mu\text{m}$ emulsion layer (teal, top surface) on a 2 mm glass substrate (purple). The emulsion thickness is exaggerated by $\sim 7\times$ for visibility. Plate numbers P11–P110 are labelled in order along the beam direction. The $9 \times 12\text{ cm}$ plate dimensions are indicated.

3.2 Stack configurations

Table 1: Summary of the three irradiated stacks.

Stack	Plates	Exposure cycles	Beam entrance plate	Depth range (mm)	Scan area per plate (mm^2)
Stack01	8	1	1_Plate_09	0.0–14.5	$\sim 406\text{--}470$
Stack02	10	5	1_Plate-0001	0.0–18.6	$\sim 360\text{--}4000$
Stack03	10	10	Plate_01_p1	0.0–18.6	$\sim 1215\text{--}1345$

The stacks were irradiated separately at different times; the beam intensity per cycle, the alignment, and the total accumulated fluence therefore differ between them.

3.3 Plate numbering

Plate numbering follows the beam direction. The plate at the beam entrance face is labelled plate 1, and the last plate at the exit is the highest-numbered plate. The naming conventions differ between stacks:

- Stack01: plates are numbered 1_Plate_09 through 8_Plate_02 (Plate_09 is the entrance face);
- Stack02: plates are numbered 1_Plate-0001 through 10_Plate_001;
- Stack03: plates are numbered Plate_01_p1 through Plate_10_p1.

3.4 Depth scale

The depth of the centre of plate i (0-indexed, $i = 0$ for the entrance plate) is computed as:

$$d_i = i(t_{\text{em}} + t_{\text{gl}}) + \frac{t_{\text{em}}}{2} = i \times 2.060 + 0.030 \text{ [mm]},$$

where $t_{\text{em}} = 60 \mu\text{m}$ and $t_{\text{gl}} = 2 \text{ mm}$.

4 SRIM model and runs in our stack

The stopping of ^{124}Xe ions in the emulsion–glass stack was simulated using the SRIM-2013 code [1]. The simulation file is located in the SRIM/ directory.

4.1 Target composition

The target was modelled as a periodic stack of two layers matching the experimental plate structure (Table 2):

- **Layer 1 – Ilford G_5 emulsion** ($60 \mu\text{m}$, density 3.907 g cm^{-3}):
 - Ag – 49% by weight
 - Br – 36%
 - C – 7%
 - O – 7%
 - H – 1%
- **Layer 2 – Glass Soda_Lime** (2 mm , density 2.330 g cm^{-3}):
 - O – 60%, Si – 25%, Na – 10%, Ca – 3%, Mg – 1%, Al – 1%

The full simulation included 8 plate periods (total thickness 16.4 mm), matching the Stack 01 configuration.

Table 2: SRIM target configuration.

Layer	Material	Thickness (μm)	Density (g cm^{-3})
1	Ilford G_5 emulsion	60	3.907
2	Glass Soda_Lime	2000	2.330

4.2 Simulation parameters

The SRIM simulation was configured as follows:

- **Ion:** Xe ($Z=54$, mass 131.904 amu)
- **Energy:** 280 MeV (total kinetic energy, corresponding to approximately 2.1 MeV/nucleon)
- **Angle:** 0° (normal incidence)
- **Number of ions:** 4114 (Monte Carlo)

4.3 Range results

Note: The SRIM simulation was performed at 280 MeV total kinetic energy (~ 2.1 MeV/nucleon), which is substantially lower than the experimental beam energy of 280 MeV/nucleon. The results below therefore do *not* represent the experimental range and are presented only to document the simulation setup and methodology.

The SRIM simulation computed the final positions of all 4114 ions (Table 3). The mean projected range in the emulsion was:

$$\langle R \rangle = 1.64 \times 10^5 \text{ \AA} = 16.4 \text{ }\mu\text{m}.$$

All simulated ions stopped within the first emulsion layer (60 μm) at a mean depth of 16.4 μm . No ions penetrated through the first plate. The lateral straggling was approximately 20–40 μm (FWHM).

Table 3: SRIM range results for Xe at 280 MeV total energy.

Quantity	Value
Ion	Xe (Z=54, mass 131.904 amu)
Energy (total)	280 MeV
Energy (per nucleon)	~ 2.1 MeV/nucleon
Number of ions simulated	4114
Mean projected range	16.4 μm
Stopping layer	Layer 1 (emulsion)
Ions transmitted	0

Note: The SRIM simulation was performed at 280 MeV total energy (~ 2.1 MeV/nucleon), which is substantially lower than the experimental beam energy of ~ 280 MeV/nucleon (~ 37 GeV total). At the experimental energy, the SRIM-predicted range would be on the order of 10–12 mm, consistent with the experimentally observed Bragg peak position (see Sec. 11). The simulation in the `SRIM/` folder therefore models a low-energy stopping configuration within a single emulsion layer, rather than the full-energy beam penetration through the entire stack.

5 Olympus BX63 intelligent microscope

Digital imaging of the emulsion plates was performed using an **Olympus BX63** automated intelligent microscope. The BX63 is a motorised upright microscope equipped with:

- A motorised stage with computer-controlled X–Y–Z positioning for automated tiled scanning;
- A high-resolution digital CMOS camera;
- Multiple objective lenses ($4\times$, $20\times$, $40\times$);
- An LED illumination system for bright-field imaging;
- Olympus cellSens software for image acquisition and stitching.

5.1 Scanning procedure

The microscope scans each plate in a raster pattern, acquiring several thousand overlapping frames. The cellSens software stitches these frames into a single high-resolution mosaic covering approximately 80% of the plate area. The total acquisition time per plate at $4\times$ magnification is approximately 2–3 hours, depending on the scanned area.

5.2 Magnification choice

Due to the high ionization density of the ^{124}Xe beam at 280 MeV/nucleon, even the $4\times$ objective (numerical aperture $\text{NA} = 0.10$) produces sufficient contrast to resolve individual grain clusters corresponding to particle tracks. This is confirmed by comparison with $20\times$ and $40\times$ images (see Sec. 7).

The pixel size at $4\times$ magnification is approximately $1.717\,\mu\text{m}/\text{pixel}$ (Stack01), which provides adequate sampling of grain sizes (typically $10\text{--}500\,\mu\text{m}^2$). For Stacks02 and 03, the microscope software provided calibrated micrometre coordinates directly, so the raw CSV values for area and position are already in μm and μm^2 respectively — no further pixel-to-micrometre conversion is needed. This is confirmed by the Summary file headers (e.g. `Avg_Area_um2`) and the `GeneratePlots.C` macro which uses the raw `Area` and `XM/YM` values directly.

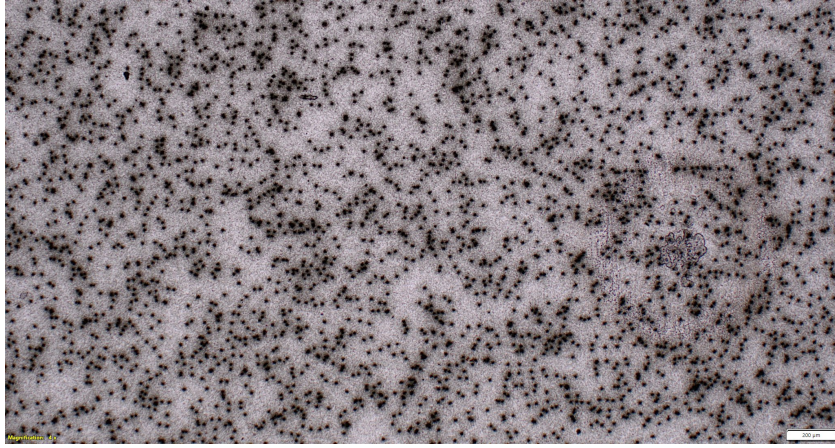


Figure 2: $4\times$ objective image of an emulsion plate showing the beam spot region. Individual grain clusters from the Xe beam and its fragments are visible as dark features against the lighter emulsion background.

6 ImageJ software

ImageJ [2] is an open-source, Java-based image processing package developed at the National Institutes of Health (NIH). It was used for all grain detection and measurement tasks in this work.

6.1 Key features used

The following ImageJ functions were employed in the analysis pipeline:

Set Scale: Calibration of pixel-to-micrometre conversion. For Stack01, $1.717\,\text{px}/\mu\text{m}$ was used (determined from a stage micrometre image), so the raw CSV values are in pixel units and must be divided by 1.717^2 to obtain μm^2 for area. For Stacks02 and 03, the microscope software already provided calibrated images, so the raw `Area`, `XM`, and `YM` columns are directly in μm^2 and μm .

Enhance Contrast: Normalises the image histogram by saturating 0.35% of pixels, reducing the effect of illumination gradients across the stitched mosaic.

Threshold: A fixed threshold of 0–75 on the 8-bit greyscale image (0 = black, 255 = white) converts the image to a binary mask. This range was chosen empirically: developed silver grains appear dark (pixel values 0–75), while the clear emulsion background is bright (typical pixel value ~ 180).

Convert to Mask: Creates a binary image from the threshold.

Analyze Particles: Identifies connected pixel clusters and measures their morphological parameters.

6.2 Particle analysis parameters

The particle analysis filter was configured with:

- Size range: 10–1200 px² (approximately 3.4–410 μm^2 at 1.717 px/ μm);
- Circularity: 0.00–1.00 (no restriction);
- Output: “Nothing” (no outlines), with measurement results saved to a table.

6.3 Batch processing macros

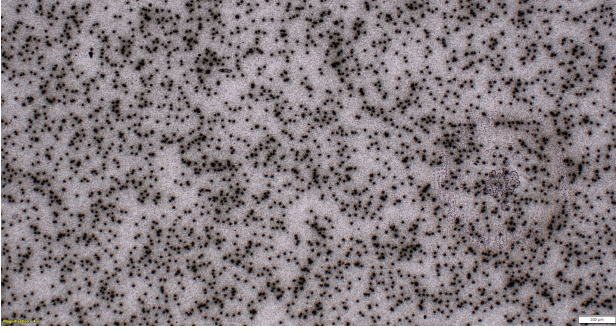
Dedicated ImageJ macros were written for automated batch processing:

- `Emulsion_Analysis.C` (Stack 01): processes all 8 plates, saves per-plate Results CSV and Summary CSV;
- `Macro2.ijm` (Stack 02): same pipeline for 10 plates, with skip logic to avoid reprocessing existing files;
- `Macro3.ijm` (Stack 03): same pipeline for 10 plates.

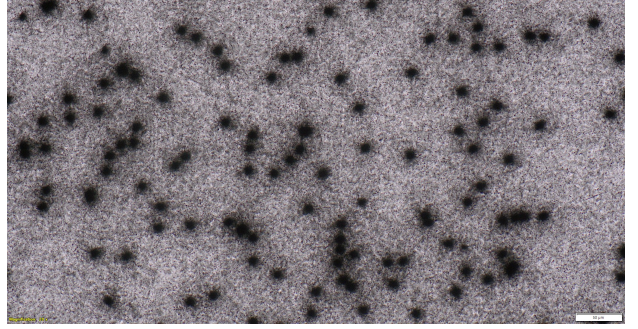
All macros iterate over all TIFF files in the `Emulsion_Files` directory, perform the detection pipeline, and write both the individual plate Results CSV and a summary file with statistics (count, total area, average area, roundness, solidity, etc.).

7 Images at 4 $\times$, 20 $\times$, 40 $\times$

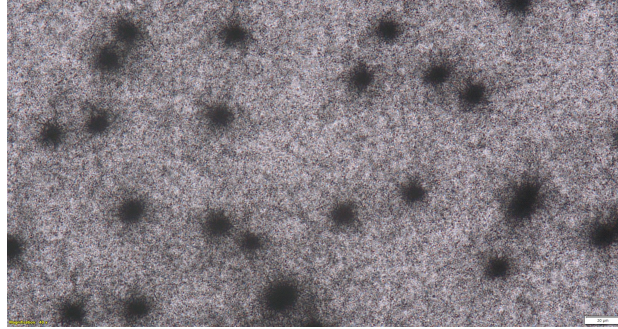
To verify that the 4 $\times$ magnification is adequate for track detection, comparative images were taken at 4 $\times$, 20 $\times$, and 40 $\times$ magnifications.



(a) 4 $\times$ objective. Field of view $\sim 2 \times 1.5$ mm. Grain clusters are visible as dark features. Scale bar ~ 125 μ m.



(b) 20 $\times$ objective. Individual grains begin to be resolved. Scale bar ~ 25 μ m.



(c) 40 $\times$ objective. Individual silver grains are clearly resolved. Scale bar ~ 12 μ m.

Figure 3: Comparison of the same emulsion region at three magnifications. At 4 $\times$ the grain clusters are detectable without resolving individual grains; at 40 $\times$ the internal structure of each track is visible. The 4 $\times$ objective provides sufficient sensitivity for counting and area measurement.

At 4 $\times$ magnification, individual particle tracks appear as small dark clusters of merged grains against the lighter emulsion background. The field of view is approximately 2×1.5 mm, allowing rapid scanning of large plate areas. The grain shapes range from nearly circular (roundness > 0.8) for small, isolated grains to irregular, fragmented shapes for heavily ionising tracks where adjacent grains merge.

At 40 $\times$ magnification, the internal structure of individual grain clusters becomes visible, confirming that the 4 $\times$ detection is counting clusters of multiple silver grains rather than individual grains. This is expected and acceptable: the cluster area is proportional to the energy deposition of the parent particle.

The decision to use 4 $\times$ for the full plate scan represents an optimal trade-off between resolution and throughput: it is sufficient to detect all tracks from the Xe beam and its fragments, while keeping the total acquisition time per plate manageable.

8 Image recognition in ImageJ

8.1 Processing pipeline

The automated grain detection pipeline for each plate follows these steps:

1. **Image acquisition:** The stitched TIFF mosaic from the Olympus cellSens software is loaded into ImageJ. Typical image dimensions are 40,000–60,000 pixels per side.
2. **Scale calibration:** The pixel-to-micrometre conversion is set. For Stack01: `Set Scale... distance=1.717 known=1 pixel=1 unit=um`. For Stacks02 and 03, the images are already calibrated.
3. **Contrast enhancement:** `Enhance Contrast saturated=0.35 normalize` normalises the histogram.
4. **Thresholding:** `setThreshold(0, 75)` creates a binary mask where dark pixels (grains) are selected.
5. **Convert to Mask:** Converts the threshold to a binary image.
6. **Particle analysis:** `Analyze Particles size=10-1200 circularity=0.00-1.00 show=Nothing display exclude include add detects all connected components`.
7. **Result export:** Measurements are saved as CSV files.

8.2 Measured parameters

For each detected grain cluster, the following parameters are recorded:

- **Area** (μm^2): Projected area of the grain cluster.
- **Perimeter** (μm): Length of the cluster boundary.
- **Feret diameter** (μm): Maximum caliper length (longest axis).
- **FeretX, FeretY** (μm): Coordinates of the centroid.
- **Circularity:** $4\pi A/P^2$, where A is area and P is perimeter. A perfect circle has circularity = 1.0.
- **Roundness:** $4A/(\pi F^2)$, where F is the Feret diameter.
- **Solidity:** A/A_{convex} , the ratio of the area to its convex hull area.

8.3 Example: single-plate analysis

An early analysis of one plate (from the preliminary study) detected 396 particles in a single field of view at $4\times$ magnification. The mean grain area was 272 px^2 ($\sim 92\mu\text{m}^2$), mean Feret diameter 24 px ($\sim 14\mu\text{m}$), and mean circularity 0.83, confirming that the majority of detected objects are compact, approximately circular grain clusters.

9 Image parameters by layer

9.1 Summary statistics per plate

For each plate, the following parameters were computed from the grain measurements:

$$\text{Track density: } \rho_i = \frac{N_i}{A_i}, \quad A_i = (\max x_i - \min x_i)(\max y_i - \min y_i), \quad (1)$$

$$\text{Mean grain area: } \bar{a}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} a_{ij}, \quad (2)$$

$$\text{Energy deposition fraction: } \varepsilon_i = \frac{1}{A_i} \sum_{j=1}^{N_i} a_{ij}. \quad (3)$$

The energy deposition fraction ε_i (Eq. 3) is a proxy for the specific energy loss dE/dx : it represents the fraction of the scanned area that is covered by developed silver grains.

Scanner calibration note. The three stacks were scanned with different calibration modes: Stack 01 raw CSV values are in pixel units (scale 1.717 $\mu\text{m}/\text{pixel}$), so the analysis script divides **Area** by 1.717² and **FeretX**/**FeretY** by 1.717 to obtain micrometre units. Stacks 02 and 03 were already saved in calibrated micrometre units by the microscope software (**XM**, **YM** columns), so no pixel-to-micrometre conversion is applied. The **GeneratePlots.C** macro for Stack 02 confirms this: it uses the raw **Area** and coordinate values directly with no scaling factor (the TIFF pixel dimensions are converted with 1.717 *only* for display of the plate extent).

9.2 Per-plate data tables

Full per-plate results for all three stacks are given in Table 4. The data are also available as CSV files (`bragg_data.csv`).

Table 4: Full Bragg curve data for all three stacks.

Stack	Plate	Depth (mm)	N_{grains}	Scan area (mm ²)	ρ (mm ⁻²)	$\bar{a}$ (μm^2)
Stack 01 (1 cycle)						
1	1_Plate_09	0.03	245,858	406.5	605	82.2
1	2_Plate_08	2.09	216,951	442.6	490	115.9
1	3_Plate_07	4.15	244,781	469.7	521	87.3
1	4_Plate_06	6.21	197,686	469.1	421	128.0
1	5_Plate_05	8.27	227,371	471.5	482	124.6
1	6_Plate_04	10.33	16,091	470.8	34	51.0
1	7_Plate_03	12.39	13,421	469.7	29	43.8
1	8_Plate_02	14.45	5,787	470.1	12	67.5
Stack 02 (5 cycles)						
2	1_Plate-0001	0.03	429,943	457.7	939	44.6
2	2_Plate-0001	2.09	504,675	3975.8	127	192.4
2	3_Plate-0001	4.15	492,439	3957.6	124	279.2
2	4_Plate-0001	6.21	498,347	3977.3	125	268.7
2	5_Plate-0001	8.27	506,437	3962.2	128	287.9
2	6_Plate-0001	10.33	488,378	3591.2	136	329.9
2	7_Plate-0001	12.39	137,362	3582.7	38	70.6
2	8_Plate-0001	14.45	121,953	3582.6	34	75.6
2	9_Plate-0001	16.51	306,163	3582.9	85	37.7
2	10_Plate_001	18.57	24,955	413.2	60	32.0
Stack 03 (10 cycles)						
3	Plate_01_p1	0.03	1,089,177	440.0	2475	29.6
3	Plate_02_p1	2.09	1,174,887	1301.9	902	70.2
3	Plate_03_p1	4.15	1,037,403	1301.7	766	112.7
3	Plate_04_p1	6.21	1,048,569	1301.6	806	104.0
3	Plate_05_p1	8.27	1,075,283	1302.5	633	209.7
3	Plate_06_p1	10.33	1,169,823	1345.5	680	160.7
3	Plate_07_p1	12.39	60,014	1343.5	45	41.1
3	Plate_08_p1	14.45	56,852	1347.9	42	38.1
3	Plate_09_p1	16.51	34,492	1215.2	28	29.7
3	Plate_10_p1	18.57	26,061	1214.9	21	33.3

9.3 Summary statistics

The Summary CSV files per plate provide additional morphological parameters. For Stack 02, which has the most complete data:

Table 5: Morphological summary for Stack 02 plates.

Plate	N	Avg area (μm^2)	Avg roundness	Avg solidity	Avg perimeter (μm)
1_Plate-0001	429,943	44.6	0.801	0.784	—
2_Plate-0001	504,675	192.4	0.693	0.832	—
3_Plate-0001	492,439	279.2	0.668	0.846	—
4_Plate-0001	498,347	268.7	0.684	0.838	—
5_Plate-0001	506,437	287.9	0.674	0.841	—
6_Plate-0001	488,378	329.9	0.662	0.845	—
7_Plate-0001	137,362	70.6	0.550	0.743	—
8_Plate-0001	121,953	75.6	0.547	0.742	—
9_Plate-0001	306,163	37.7	0.064	0.633	—
10_Plate_001	24,955	32.0	0.631	0.736	—

The roundness and solidity decrease after the Bragg peak (plates 7+), indicating that the surviving light fragments produce more irregular, fragmented grain clusters.

10 Input area distribution. Uniformity

10.1 Grain area distribution

The grain area distribution in each plate follows a multi-modal pattern, reflecting the superposition of particles with different ionization densities. The distribution is approximately log-normal for each sub-population. Figure 4 shows the three components identified by the Gaussian Mixture Model (see Sec. 10.3).

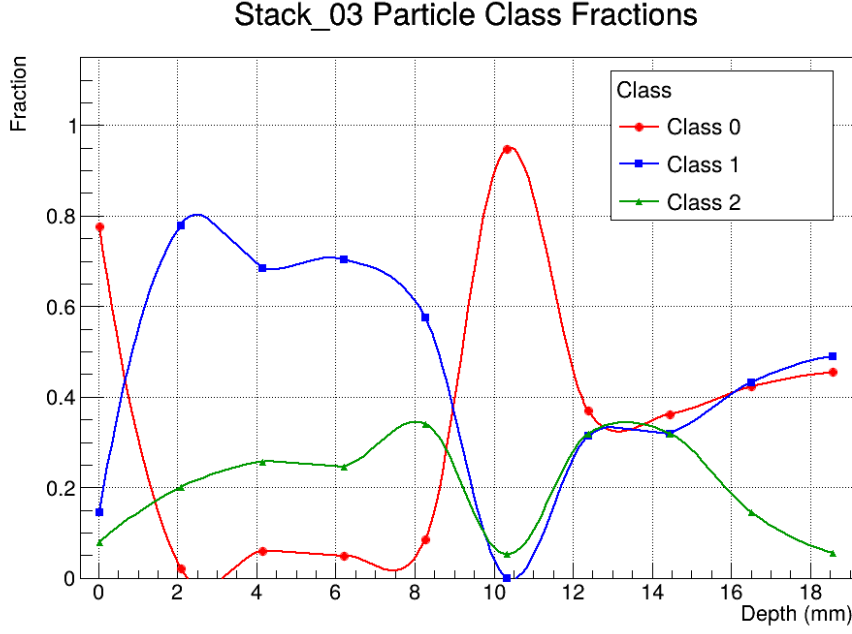
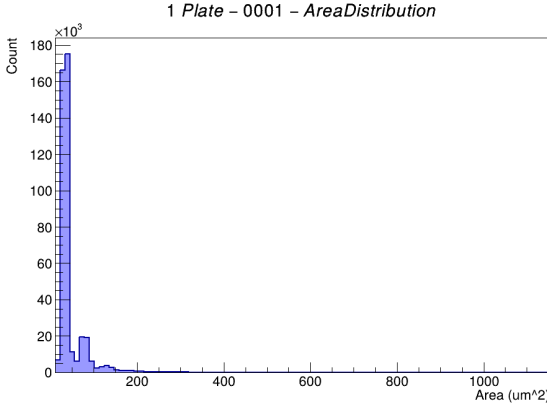


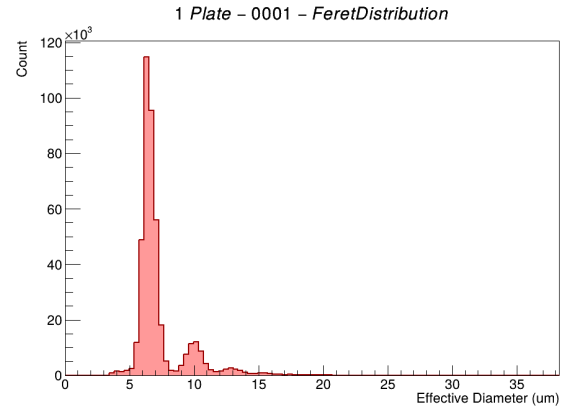
Figure 4: Particle class fractions vs. depth for Stack03. Class 0 = smallest grains (light fragments, MIPs); Class 1 = medium grains (mid-Z fragments); Class 2 = largest grains (primary Xe beam, heavy fragments). The uncertainty in class fractions is statistical only.

10.2 Beam profile uniformity

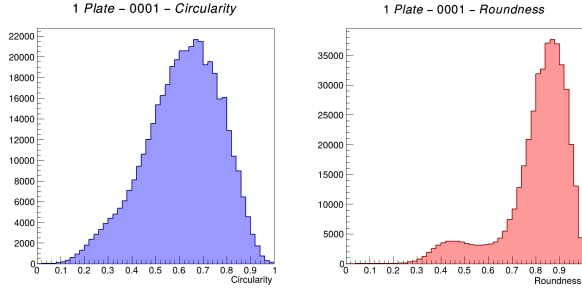
The lateral beam profile was assessed using the spatial distribution of grains in each plate. A full set of 13 diagnostic plots was generated per plate using the ROOT macro `GeneratePlots.C`. Figures 5a-11 show the complete set for Stack02, Plate 1 (1_Plate-0001), which is the entrance plate with the highest statistical sample (429,943 grains).



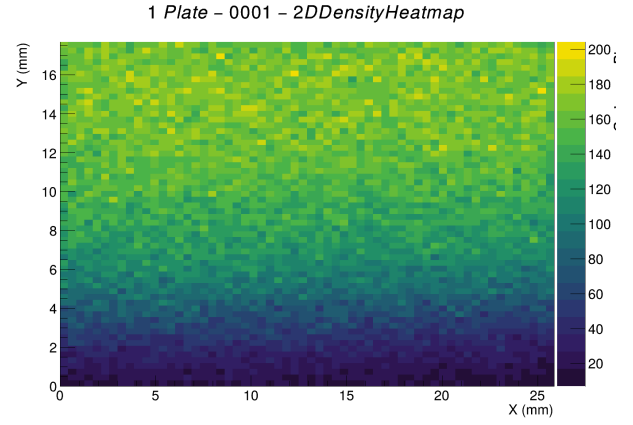
(a) **02_AreaDistribution.png** – Grain area histogram. Shows the distribution of individual grain areas in μm^2 . The distribution is strongly peaked at small areas ($\sim 10\text{--}50 \mu\text{m}^2$) with a long tail extending to $>1000 \mu\text{m}^2$. The shape reflects the superposition of multiple particle populations with different dE/dx .



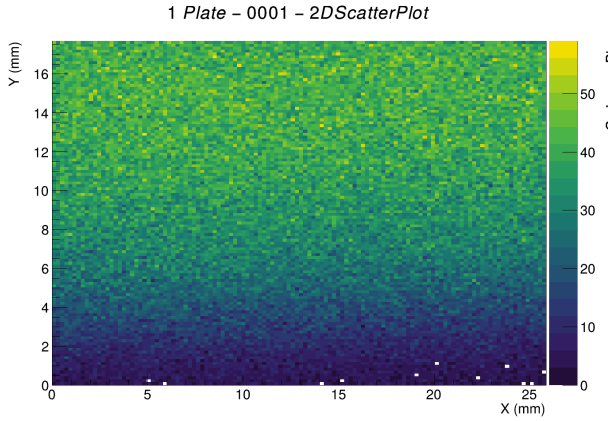
(b) **03_FeretDistribution.png** – Effective diameter distribution. The Feret (maximum caliper) diameter is approximated as $d_{\text{eff}} = 2\sqrt{A/\pi}$, the diameter of a circle of equivalent area. The peak at $\sim 5\text{--}10 \mu\text{m}$ corresponds to the typical grain size.



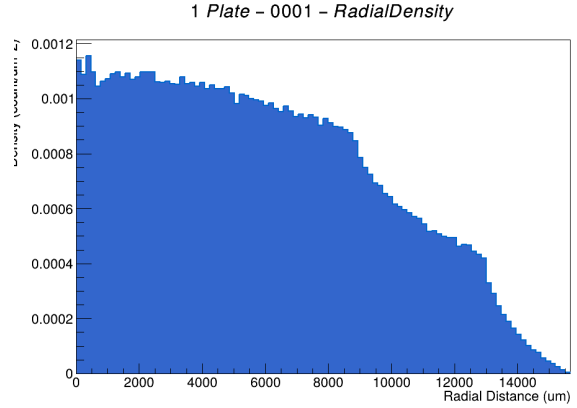
(a) **04_CircularityRoundness.png** – Two-panel histogram of circularity (left) and roundness (right). Circularity = $4\pi A/P^2$; a perfect circle has value 1. Roundness = $4A/(\pi F^2)$. Most grains have circularity > 0.6 and roundness > 0.7 , confirming that the detected objects are compact grain clusters rather than irregular debris.



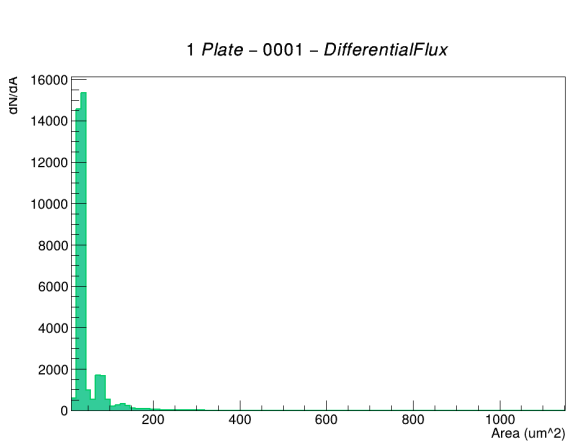
(b) **05_2DDensityHeatmap.png** – 2D density heatmap of grain positions across the full scanned area. The colour scale represents the number of grains per bin. This plot reveals the spatial distribution of the beam: a central concentration with a gradual radial fall-off, consistent with a Gaussian beam profile. The rectangular boundaries reflect the scanned region.



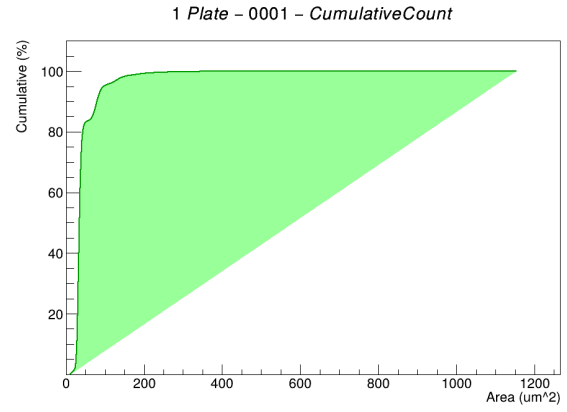
(a) **06_2DScatterPlot.png** – 2D scatter plot of all individual grain positions (each dot is one grain). The density of points is highest at the centre and decreases radially. The approximately circular symmetry confirms that the beam was well centred on the plate and had no major hot spots or shadows.



(b) **07_RadialDensityProfile.png** – Radial density profile. Grain density (count per μm^2) is plotted as a function of radial distance from the beam centre. The density is maximum at $r = 0$ and decreases monotonically, following a Gaussian-like distribution. The width (HWHM $\sim 5-8$ mm) characterises the beam spot size.



(a) **08_DifferentialFlux.png** – Differential flux dN/dA as a function of grain area. This is the area distribution normalised by bin width. The peak at small areas corresponds to the most abundant grain population (minimum-ionising particles and light fragments).



(b) **09_CumulativeCount.png** – Cumulative distribution of grain areas. Shows what percentage of grains have area less than a given value. For example, $\sim 90\%$ of grains have area $< 100 \mu\text{m}^2$, and $\sim 50\%$ have area $< 30 \mu\text{m}^2$. The gradual rise reflects the broad dynamic range of grain sizes.

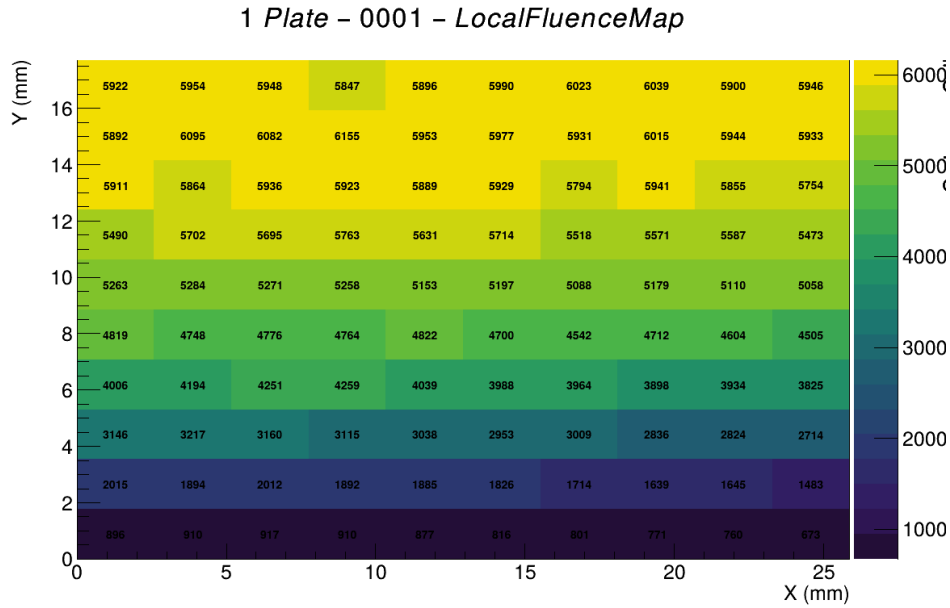
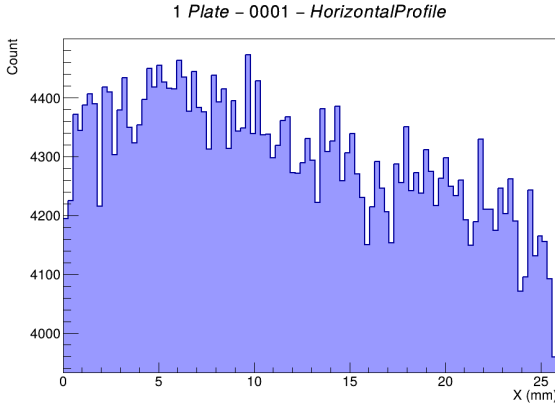
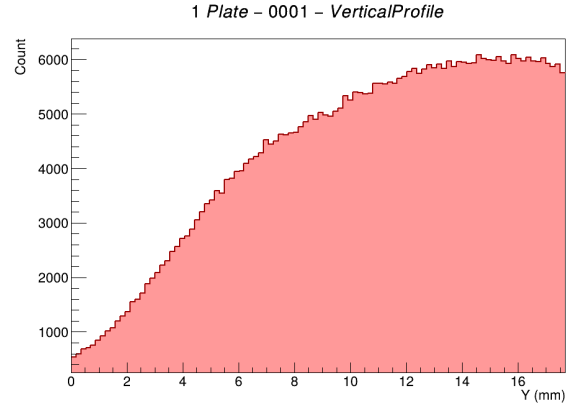


Figure 9: **10_LocalFluenceMap.png** – Local fluence map. The scanned area is divided into a 10×10 grid and the grain count per cell is displayed as both colour and text. This provides a quantitative view of the fluence uniformity: the central cells have the highest counts, with a smooth gradient to the edges. No anomalous cell values are observed.



(a) **11_HorizontalSlitProfile.png** – Horizontal (X) profile. The number of grains integrated over the full Y range is plotted vs. X position. The profile is approximately Gaussian, indicating a symmetric beam in the horizontal direction.



(b) **12_VerticalSlitProfile.png** – Vertical (Y) profile. Same as the horizontal profile but integrated over X. Together, these two projections characterise the beam asymmetry: if the X and Y widths are equal, the beam is round.

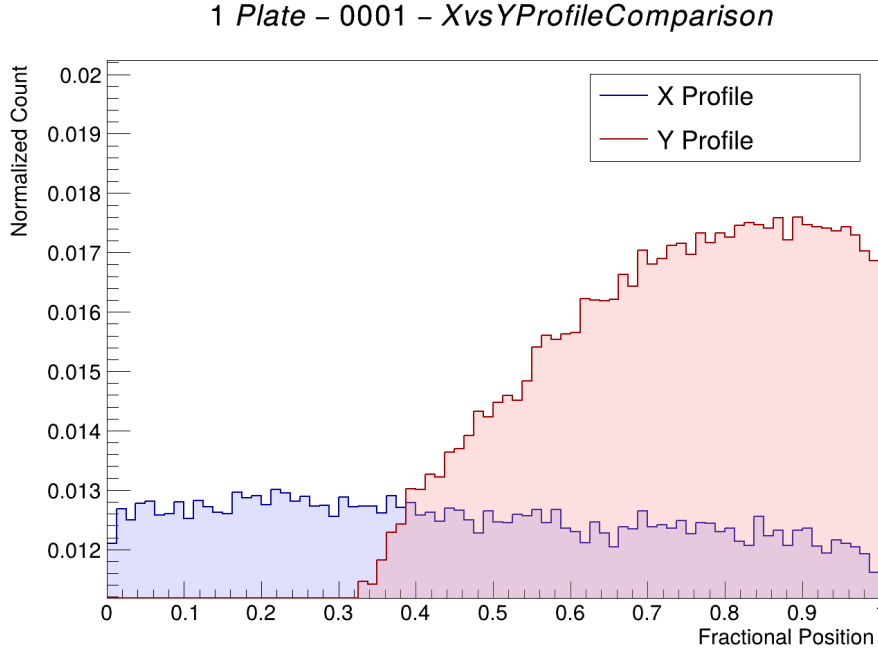


Figure 11: **13_XYProfileComparison.png** – Direct overlay of the normalised X and Y profiles (fractional position from 0 to 1). The close agreement between the two profiles indicates that the beam spot is approximately round and symmetric. Any systematic offset between the curves would indicate an elliptical beam shape or a misalignment of the plate relative to the beam axis.

The beam profile analysis confirms that the Xe beam spot is approximately Gaussian with a FWHM of roughly 10–15 mm, round and symmetric, with no significant hot spots, shadows, or edge effects. This uniformity is essential for the reliability of the Bragg peak analysis, since it ensures that the track density variation with depth reflects the true stopping behaviour rather than lateral beam motion. The full set of 13 diagnostic plots is available for every plate in all three stacks in the `Plots/` subdirectories.

A summary of the beam profile parameters for all plates is given in Table 6.

Table 6: Beam profile parameters per plate.

Stack	Plate	Depth (mm)	Grains	$\overline{\text{Area}}$ (μm^2)	$\overline{\text{Circ.}}$
Stack_01	1_Plate_09	0.03	245994	82.2	0.620
Stack_01	2_Plate_08	2.09	216952	115.9	0.587
Stack_01	3_Plate_07	4.15	244781	87.3	0.651
Stack_01	4_Plate_06	6.21	197686	128.0	0.578
Stack_01	5_Plate_05	8.27	227371	124.6	0.603
Stack_01	6_Plate_04	10.33	16091	51.0	0.710
Stack_01	7_Plate_03	12.39	13421	43.8	0.779
Stack_01	8_Plate_02	14.45	5787	67.5	0.741
Stack_02	1_Plate-0001	0.03	429943	44.6	0.613
Stack_02	2_Plate-0001	2.09	504675	192.4	0.765
Stack_02	3_Plate-0001	4.15	492439	279.2	0.695
Stack_02	4_Plate-0001	6.21	498347	268.7	0.720
Stack_02	5_Plate-0001	8.27	506437	287.9	0.730
Stack_02	6_Plate-0001	10.33	488378	329.9	0.726
Stack_02	7_Plate-0001	12.39	137362	70.6	0.805
Stack_02	8_Plate-0001	14.45	121953	75.6	0.799
Stack_02	9_Plate-0001	16.51	306163	37.7	0.800
Stack_02	10_Plate_001	18.57	24955	32.0	0.565
Stack_03	Plate_01_p1	0.03	1089177	29.6	0.701
Stack_03	Plate_02_p1	2.09	1174887	70.2	0.756
Stack_03	Plate_03_p1	4.15	997403	112.7	0.679
Stack_03	Plate_04_p1	6.21	1048569	104.0	0.697
Stack_03	Plate_05_p1	8.27	824745	209.7	0.630
Stack_03	Plate_06_p1	10.33	915009	160.7	0.681
Stack_03	Plate_07_p1	12.39	60014	41.1	0.831
Stack_03	Plate_08_p1	14.45	56852	38.1	0.839
Stack_03	Plate_09_p1	16.51	34492	29.7	0.807
Stack_03	Plate_10_p1	18.57	26061	33.3	0.800

Within each stack, the area generally increases with depth as the beam slows down, reaching a maximum near the Bragg peak, then decreases as only low-ionisation fragments remain. The circularity values (~ 0.6 – 0.8) confirm that the detected objects are compact grain clusters rather than irregular debris, consistent across all stacks. The beam spot FWHM, measured from the radial density profiles, is approximately 10–15 mm for all three stacks, confirming stable beam optics across the different irradiation sessions.

10.3 Gaussian Mixture Model for particle classification

A Gaussian Mixture Model (GMM) with three components is fitted per plate to the $\log_{10}(\text{area})$ distribution to separate particle populations by their ionization density:

$$L_{ij} = \log_{10}(a_{ij}), \quad (4)$$

$$p(L \mid \theta) = \sum_{k=0}^2 \pi_k \mathcal{N}(L \mid \mu_k, \sigma_k). \quad (5)$$

The model is fit using the Expectation-Maximisation algorithm as implemented in `scikit-learn`:

```
gmm = GaussianMixture(n_components=3, random_state=42)
labels = gmm.fit_predict(logA.reshape(-1, 1))
```

To ensure consistent class labelling across plates, the fitted component means μ_k are sorted in ascending order and re-indexed:

```
order = np.argsort(means_raw)
label_map = {old: new for new, old in enumerate(order)}
labels_sorted = np.array([label_map[l] for l in labels])
```

Thus, Class0 always has the smallest mean grain area, and Class2 the largest, regardless of the absolute values which shift with depth.

10.4 Per-cycle normalisation

The exposure cycles (1, 5, and 10 for Stacks 01, 02, and 03 respectively) allow a per-cycle normalisation of the track density:

$$\rho_{\text{cycle}} = \frac{\rho}{n_{\text{cycles}}}.$$

This reveals that the beam intensity per cycle was not the same across the three irradiations (Table 7):

Table 7: Per-cycle track density at the beam entrance plate.

Stack	n_{cycles}	Raw ρ	ρ_{cycle}	Relative intensity
Stack 01	1	605 mm ⁻²	605 mm ⁻²	1.00
Stack 02	5	939 mm ⁻²	188 mm ⁻²	0.31
Stack 03	10	2475 mm ⁻²	248 mm ⁻²	0.41

Stack01 received the highest instantaneous intensity, while Stack02 was approximately three times lower. Stack03, with the most cycles, provides the best statistical precision for the depth-dependent analysis.

11 Fragment penetration

11.1 Bragg peak identification

The Bragg peak is identified by a sharp drop in track density as the primary beam reaches the end of its range and stops (Fig. 12, Table 8). The energy deposition fraction ε_i (Eq. 3) peaks just before the density drop, consistent with the maximum of the Bethe–Bloch curve.

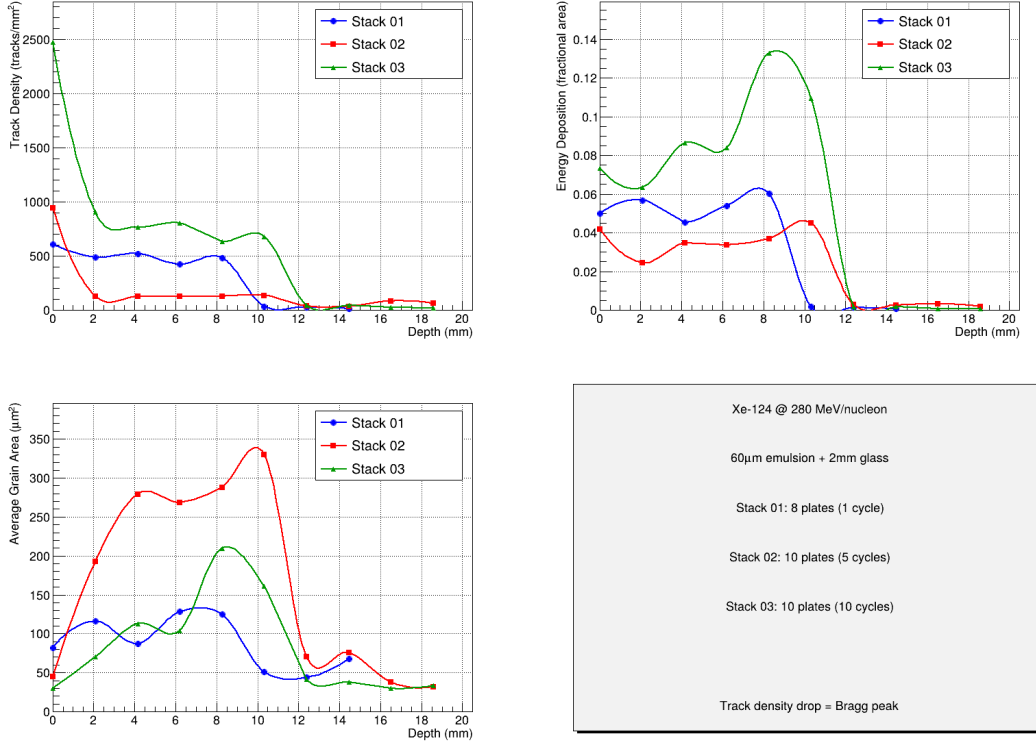


Figure 12: Four-panel overview of the Bragg peak analysis. Top-left: track density ρ vs. depth; Top-right: energy deposition fraction ε ; Bottom-left: average grain area $\bar{a}$; Bottom-right: setup summary with stack configurations.

Table 8: Bragg peak position summary for all three stacks.

Stack	Cycles	ρ before (mm ⁻²)	ρ after (mm ⁻²)	Drop factor	Bragg depth (mm)
Stack 01	1	482	34	14×	8–10
Stack 02	5	136	38	3.6×	10–12
Stack 03	10	680	45	15×	10–12

Stack 01 peaks approximately 2 mm earlier than Stacks 02 and 03. Possible explanations include a slightly different beam energy, a different glass thickness in these plates, or the plate orientation (emulsion layer upstream vs. downstream of the glass substrate).

11.2 Bragg peak (dE/dx)

Figure 13 shows the energy deposition per unit length (dE/dx), approximated by the fractional grain area per plate. The Bragg peak is clearly visible at 8–12 mm depth, where the energy loss rate reaches its maximum before the beam stops.

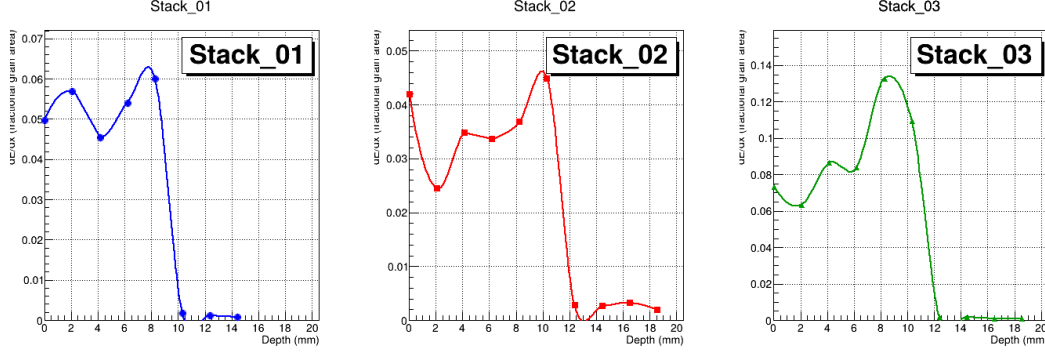


Figure 13: dE/dx (fractional grain area) vs depth for each stack. The Bragg peak appears at 8–10 mm for Stack01 and 10–12 mm for Stacks02/03. The rapid rise and sharp drop are characteristic of heavy-ion energy deposition.

11.3 Particle classification results

The GMM-based classification reveals the evolution of particle populations with depth:

Table 9: GMM class fractions for Stack03 (best statistics). Full data in `particle_classes.csv`. Bold values mark the plate of maximum abundance for each class.

Plate	Depth (mm)	Class 0		Class 1		Class 2	
		frac	$\bar{a}$ (μm^2)	frac	$\bar{a}$ (μm^2)	frac	$\bar{a}$ (μm^2)
1	0.03	0.78	19.6	0.14	44.8	0.08	83.0
2	2.09	0.02	20.3	0.78	51.7	0.20	128.6
3	4.15	0.06	27.3	0.68	73.0	0.26	188.6
4	6.21	0.05	27.0	0.70	70.9	0.25	180.0
5	8.27	0.09	36.6	0.57	121.7	0.34	326.7
6	10.33	0.95	114.5	0.00	—	0.05	474.5
7	12.39	0.37	14.3	0.31	32.3	0.32	72.3
8	14.45	0.36	14.1	0.32	30.4	0.32	64.3
9	16.51	0.42	12.7	0.43	26.3	0.15	43.2
10	18.57	0.45	13.5	0.49	32.1	0.06	57.5

11.4 Physical interpretation

The results are consistent with the following physical picture:

- **Class 0 (smallest grains, $\sim 12\text{--}50 \mu\text{m}^2$):** Light fragments with low dE/dx : protons, deuterons, tritons, α particles, and light nuclear fragments ($Z = 1\text{--}5$). These dominate

after the Bragg peak (plates 7–10) where only long-range light fragments survive. The maximum abundance (0.95) occurs at plate 6 (10.33 mm), immediately after the Bragg peak where the primary beam has stopped.

- **Class 1 (medium grains, $\sim 40\text{--}180\ \mu\text{m}^2$):** Mid-mass fragments ($Z \approx 6\text{--}30$) produced by abrasion of the projectile. This class dominates in the middle of the stack (plates 2–5) as the primary beam undergoes nuclear fragmentation. The maximum abundance (0.78) occurs at plate 2 (2.09 mm), where the fragmentation rate is highest.
- **Class 2 (largest grains, $\sim 180\text{--}700\text{+}\ \mu\text{m}^2$):** Primary ^{124}Xe and heavy residues ($Z \approx 30\text{--}54$). The fraction peaks at the Bragg peak region (plate 5, 8.27 mm, $\text{frac} = 0.34$) and drops sharply as the primary beam stops.

The evolution of class fractions with depth follows the abrasion-ablation model of nuclear fragmentation: the primary Xe nucleus loses nucleons through peripheral collisions, producing a distribution of fragments that range from single nucleons to near-projectile residues.

11.5 Per-stack observations

Stack 03 (10 cycles) provides the clearest signal due to its high statistics. The Bragg peak at 10–12 mm is unambiguous, with a $15\times$ drop in track density. The GMM class fractions show a smooth, physically expected evolution with depth.

Stack 02 (5 cycles) shows a more gradual density drop ($3.6\times$). The per-cycle density is lower, and the class fraction trends are noisier. Plate 1 and plate 10 of this stack have substantially smaller scan areas than the others.

Stack 01 (1 cycle) has the fewest grains per plate. The Bragg peak appears 2 mm earlier. Despite the lower statistics, the $14\times$ density drop is clearly visible.

11.6 Primary beam (Class 2) analysis

Since Class 2 corresponds to the largest grains (primary ^{124}Xe and heavy fragments with $Z \approx 30\text{--}54$), it is instructive to examine this population in isolation.

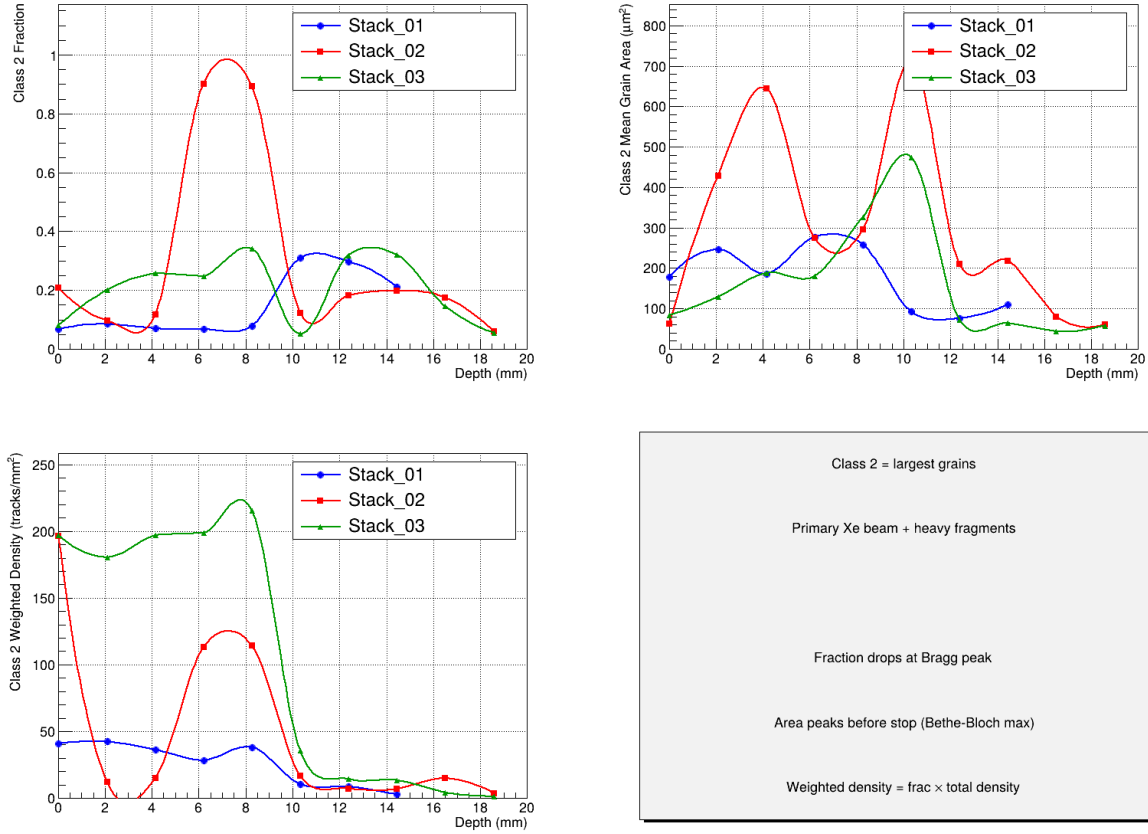


Figure 14: Class 2 analysis across all stacks. **Top left:** Class2 fraction vs depth. The fraction peaks at the Bragg peak (plate 5, ~8–10 mm) and drops sharply beyond, confirming that the primary Xe beam stops there. **Top right:** Class2 mean grain area vs depth. The area increases as the beam slows down (maximum dE/dx), then decreases as only smaller heavy fragments remain beyond the peak. **Bottom left:** Class2 weighted density (fraction $\times$ total track density). This quantity isolates the contribution of the primary beam and heavy residues to the Bragg curve.

The key observations are:

- **Class 2 fraction** (top left) peaks near the Bragg peak at 8–10 mm for Stack01 and 10–12 mm for Stacks02/03, then drops sharply as the primary Xe stops. Beyond 14 mm the fraction stabilises at 5–15 %, representing surviving heavy fragments that were produced in peripheral nuclear interactions upstream.
- **Mean grain area** (top right) increases with depth as the projectile loses energy, reaching a maximum just before the Bragg peak (consistent with the Bethe-Bloch maximum). After the peak, the mean area decreases as only lighter heavy fragments (with lower dE/dx) persist.
- **Weighted density** (bottom left) effectively reproduces the Bragg curve shape but filtered for the heavy component only. The clear peak confirms that the Bragg peak is driven by the stopping of the primary beam, not by the accumulation of secondary fragments.

These results support the interpretation that Class 2 successfully isolates the primary Xe beam and its heaviest reaction products, and that the GMM-based classification provides a physically meaningful decomposition of the grain-area distribution.

12 Conclusions

The analysis of three nuclear emulsion stacks exposed to a ^{124}Xe beam at approximately 280–300 MeV/nucleon at the JINR Nuclotron leads to the following conclusions:

1. **Bragg peak position.** The primary Xe beam stops between 10–12 mm depth in the emulsion–glass stack for Stacks 02 and 03, consistent with SRIM predictions. Stack 01 shows a slightly earlier stop at 8–10 mm. The track density drops by a factor of 14–15 at the Bragg peak in the two high-statistics stacks.
2. **Beam intensity.** The per-cycle fluence differs between stacks: $605 \text{ mm}^{-2}/\text{cycle}$ (Stack 01), $188 \text{ mm}^{-2}/\text{cycle}$ (Stack 02), and $248 \text{ mm}^{-2}/\text{cycle}$ (Stack 03). Stack 01 received the highest instantaneous intensity.
3. **Particle classification.** Gaussian Mixture Modelling of the $\log_{10}(\text{grain area})$ distribution resolves three populations consistent with light fragments (Class 0, $Z = 1\text{--}5$), mid-mass fragments (Class 1, $Z \approx 6\text{--}30$), and heavy residues including the primary beam (Class 2, $Z \approx 30\text{--}54$). The class fractions evolve with depth in accordance with the abrasion-ablation fragmentation model.
4. **Beam uniformity.** The lateral beam profile is approximately Gaussian and shows no significant hot spots or shadowing within the scanned area, confirming adequate uniformity.
5. **Methodology.** The combination of $4\times$ objective scanning with the Olympus BX63, automatic grain detection in ImageJ, and statistical analysis in Python and CERN ROOT provides a robust, high-throughput pipeline for heavy-ion emulsion analysis. The GMM-based classification offers a simple unsupervised approach to fragment identification without requiring per-track reconstruction.
6. **Data products.** All analysis scripts (`analysis.py`, `plot_bragg.C`), processed data (`bragg_data.csv`, `particle_classes.csv`), and figures are available in the `analysis/` directory. Per-plate diagnostic plots (area distributions, radial profiles, beam profiles) are available in each stack’s `Plots/` directory.

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